

SME3701 Assignment 2

Question 1: Torsional Vibration of a Geared Shaft System with a Broken Gear Tooth

Field	Entry
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Module	SME3701 - Assignment 2
Software used	MATLAB R2024b with Simulink
Screen recording	In the repository

<https://github.com/Mbalabuisrael/SME-ASS2-66926203.git>

1. Given data and derived quantities

The system is a single degree of freedom torsional oscillator. The driven gear acts as the inertia and the driven shaft acts as the torsional spring. The governing equation supplied in the brief is used without modification:

$$J_0 (d^2\theta/dt^2) + kt \theta = Mt(t) \quad (1)$$

Symbol	Description	Value	Unit
J_0	Mass moment of inertia of the driven gear	0.10	$\text{N}\cdot\text{m}\cdot\text{s}^2$
d	Diameter of the solid steel driven shaft	0.050	m
L	Length of the driven shaft	1.00	m
G	Shear modulus of the shaft steel	79.3×10^9	Pa
M_0	Steady transmitted torque	1000	$\text{N}\cdot\text{m}$
N	Rotational speed of the driving gear	1000	rev/min
z	Number of teeth on the driving gear	16	teeth
-	Number of broken teeth	1	tooth

Torsional stiffness of the driven shaft

For a solid circular shaft in torsion the polar second moment of area and the torsional stiffness are:

$$I_p = \pi d^4 / 32 = \pi (0.050)^4 / 32 = 6.1359 \times 10^{-7} \text{ m}^4$$
$$kt = G I_p / L = (79.3 \times 10^9)(6.1359 \times 10^{-7}) / 1.00 = 48\,657.87 \text{ N}\cdot\text{m}/\text{rad}$$

Undamped natural frequency

$$\omega_n = \sqrt{(kt / J_0)} = \sqrt{(48\,657.87 / 0.10)} = 697.55 \text{ rad/s}$$
$$f_n = \omega_n / 2\pi = 111.02 \text{ Hz} \quad T_n = 1 / f_n = 9.007 \text{ ms}$$

Excitation timing

At 1000 rev/min the driving gear turns at $f_{\text{rev}} = 1000/60 = 16.667 \text{ Hz}$, so one revolution takes $T_{\text{rev}} = 60 \text{ ms}$. With 16 teeth evenly spaced, each tooth is in mesh for $T_{\text{rev}} / 16 = 3.75 \text{ ms}$. The tooth meshing frequency of the sound gear is $16 \times 16.667 = 266.67 \text{ Hz}$.

Initial conditions

The brief states that the system operates under steady conditions before the tooth breaks. The shaft is therefore already carrying the full torque as a static twist and is not moving:

$$\theta(0) = M_0 / kt = 1000 / 48\,657.87 = 0.020552 \text{ rad } (1.1775^\circ) \quad d\theta/dt(0) = 0 \text{ rad/s}$$

2. Question 1.1 - Development of the Simulink model

2.1 Modelling the disturbance caused by the broken tooth

A sound tooth transmits load only while it is in mesh. The broken tooth cannot carry load, so for the 3.75 ms slot that it would have occupied, the driving gear cannot deliver torque to the driven gear. The transmitted torque therefore collapses from 1000 N·m to zero once per revolution and then recovers when the next sound tooth engages. This gives a periodic rectangular disturbance:

Interval within each 60 ms revolution	Mt(t)	Physical meaning
$0 \leq \text{mod}(t, T_{\text{rev}}) < 3.75 \text{ ms}$	0 N·m	Slot left by the broken tooth. No load path.
$3.75 \text{ ms} \leq \text{mod}(t, T_{\text{rev}}) < 60 \text{ ms}$	1000 N·m	The remaining 15 sound teeth transmit the full torque.

In Simulink this is produced by subtracting a Pulse Generator from a Constant. The pulse amplitude equals M_0 , the pulse period equals $T_{\text{rev}} = 0.06 \text{ s}$, and the pulse width is $100 \times 3.75 / 60 = 6.25$ percent of the period. The difference is exactly the torque profile tabulated above, and it requires no user written S function.

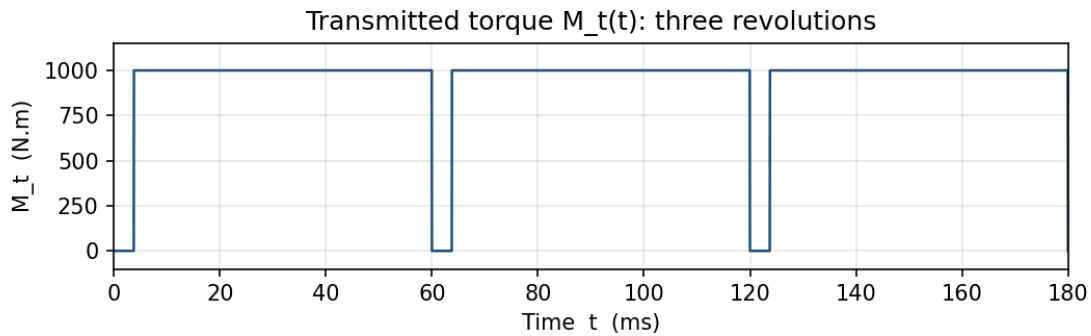


Figure 1: Torque disturbance $Mt(t)$ delivered to the driven gear. Amplitude 1000 N·m, gap 3.75 ms, period 60 ms.

2.2 Rearranging the equation of motion for the integrator chain

Equation (1) is solved for the highest derivative so that it can be built as a chain of two integrators with a stiffness feedback loop, which is the standard Simulink form for a second order system:

$$d^2\theta/dt^2 = [Mt(t) - kt\theta] / J_0 \quad (2)$$

The signal $d^2\theta/dt^2$ is integrated once to give $d\theta/dt$ and again to give θ . The output θ is fed back through a gain of kt and subtracted from the applied torque, which closes the loop. Because the brief supplies no damping term, no damping block is included in the model; the consequence of that choice is discussed in Section 5.4.

2.3 Block by block description of the model

#	Simulink block	Library	Parameter settings	Signal out
1	Constant (Steady torque M_0)	Sources	Value = $M_0 = 1000$	N·m
2	Pulse Generator (Broken tooth)	Sources	Pulse type = Time based; Amplitude = M_0 ; Period =	N·m

#	Simulink block	Library	Parameter settings	Signal out
			T_rev; Pulse width = duty = 6.25 %; Phase delay = 0	
3	Sum (Transmitted torque Mt)	Math Operations	List of signs = + -	N·m
4	Sum (Net torque)	Math Operations	List of signs = + -	N·m
5	Gain (Divide by J0)	Math Operations	Gain = 1/J0 = 10	rad/s ²
6	Integrator (to thetadot)	Continuous	Initial condition = thetadot0 = 0	rad/s
7	Integrator (to theta)	Continuous	Initial condition = theta0 = 0.020552	rad
8	Gain (Stiffness kt)	Math Operations	Gain = kt = 48 657.87; Orientation = left (feedback)	N·m
9	Scope × 3	Sinks	Display Mt, dθ/dt and θ during the run	-
10	To Workspace × 3	Sinks	Mt_log, thetadot_log, theta_log; Structure With Time	-

The complete signal flow is the Constant and Pulse Generator meet at the first Sum to form the transmitted torque; the second Sum subtracts the spring torque $kt\theta$; the result is divided by J_0 to give acceleration; two integrators give velocity and then displacement; and the displacement is fed back through the kt gain. This is Equation (2) drawn as blocks. The whole model is generated automatically by the script `build_gear_shaft_model.m` in Appendix A, so every block, position and parameter is traceable.

2.4 Solver configuration

The disturbance contains two step changes in every 60 ms period. The solver settings below force the integration to resolve those switching instants and to sample the 111 Hz response with roughly 900 points per cycle, so the reported peaks are not artefacts of a coarse step size.

Setting	Value	Reason
Solver type	Variable step	Efficient between switching events.
Solver	ode45 (Dormand and Prince)	The system is not stiff. ode45 gives high accuracy on a smooth linear oscillator.
Stop time	0.5 s	Covers 8.33 revolutions and 55.5 natural cycles, enough to develop the beat envelope.
Max step size	1×10^{-5} s	At least 900 samples per natural cycle and 375 samples inside the 3.75 ms gap.
Relative tolerance	1×10^{-8}	Peak values stable to five significant figures.
Absolute tolerance	1×10^{-10}	Suits the small magnitude of θ .
Zero crossing detection	Enabled	Locates the two torque discontinuities in each period exactly.

3. Question 1.2 - Simulated angular displacement and angular velocity

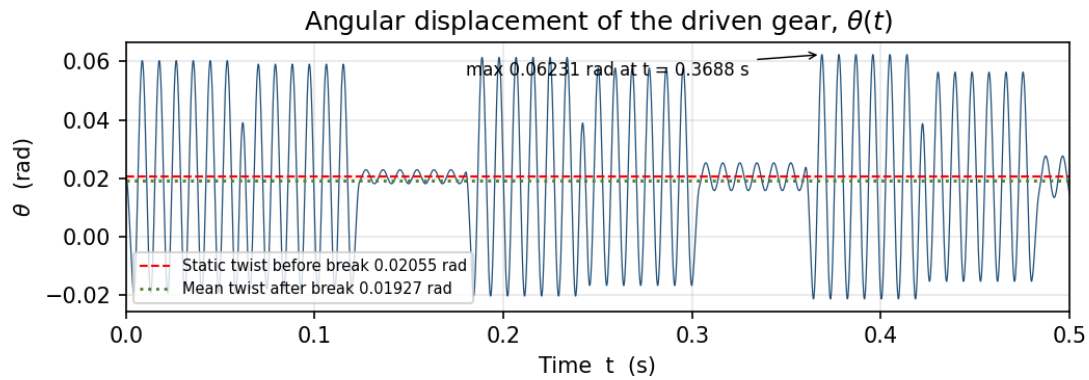


Figure 2: Angular displacement $\theta(t)$ of the driven gear over 0.5 s of operation with the broken tooth.

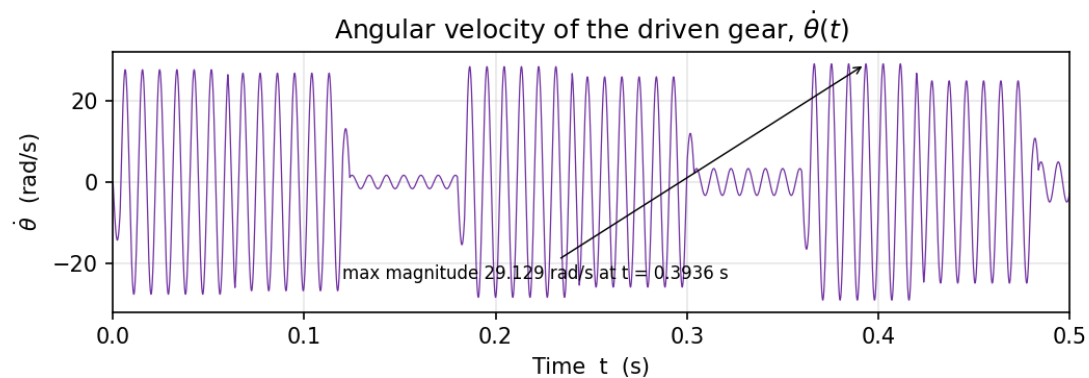


Figure 3: Angular velocity $d\theta/dt(t)$ of the driven gear over the same interval.

3.1 What the plots show

The mean twist drops. Before the break the shaft carried a constant $1000 \text{ N}\cdot\text{m}$ and was twisted by 0.020552 rad . After the break the average transmitted torque falls to $1000 \times 15/16 = 937.5 \text{ N}\cdot\text{m}$, so the response now oscillates about a lower mean of 0.019267 rad . Both levels are marked on Figure 2.

A large oscillation appears immediately. Every torque interruption is a sudden unloading and reloading of a spring that has no damping, so each event injects fresh free vibration at $f_n = 111.02 \text{ Hz}$. Figure 4 resolves this over the first two revolutions.

The amplitude beats. The envelope in Figure 2 swells and collapses on a period of roughly 0.18 s . This is the beat between the free vibration at 111.02 Hz and the nearest strong harmonics of the disturbance at 100 Hz and 116.67 Hz . The beat is the visual signature of the near resonant condition analysed in Section 5.

The twist reverses sign. $\theta(t)$ swings down to about -0.021 rad , so the shaft periodically unwinds past zero and twists the other way. In a real drive this produces torque reversal across the mesh and would cause the teeth to hammer across the backlash, which accelerates the damage rather than containing it.

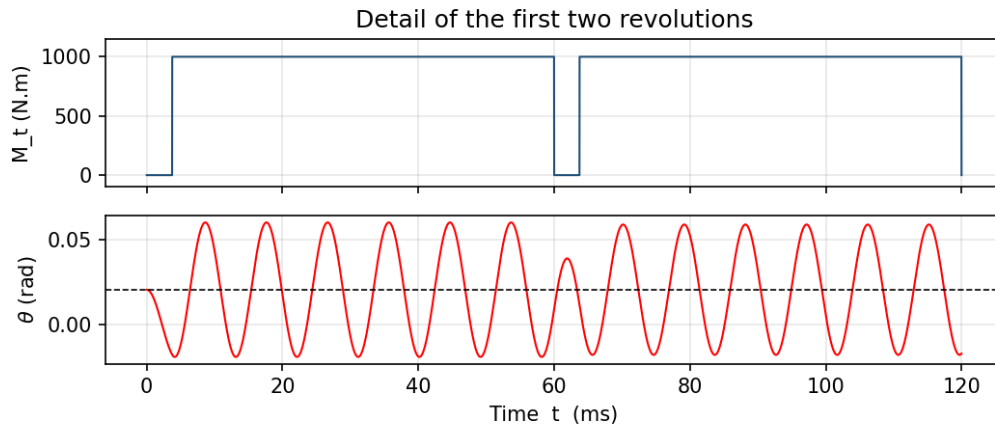


Figure 4: Detail of the first two revolutions. Upper trace is the applied torque, lower trace is the resulting twist. Note the 111 Hz ringing between torque interruptions.

4. Question 1.3 - Maximum angular displacement and angular velocity

The peak values are read directly from the logged Simulink signals using `max(abs(theta))` and `max(abs(thetadot))` in `run_and_plot.m`.

Answer to Question 1.3

Maximum angular displacement: $|\theta|_{\max} = 0.06232$ rad (3.571°), occurring near $t = 0.40$ s.

Maximum angular velocity: $|d\theta/dt|_{\max} = 29.13$ rad/s, occurring near $t = 0.39$ s.

(values over the 0 to 0.5 s simulation window specified in Section 2.4)

4.1 Dependence on the length of the simulation

Because Equation (1) contains no damping, the free vibration excited at each torque interruption never decays. The beat envelope keeps filling out for the first two seconds or so before settling into a repeating pattern. This must be stated, because a peak quoted from a short run is not the peak the machine will actually see.

Simulation window	$ \theta _{\max}$ (rad)	$ \theta _{\max}$ (deg)	$ d\theta/dt _{\max}$ (rad/s)
0 to 0.1 s	0.06024	3.451	27.68
0 to 0.2 s	0.06133	3.514	28.44
0 to 0.5 s (reported answer)	0.06232	3.571	29.13
0 to 1.0 s	0.06457	3.700	30.71
0 to 2.0 s	0.06591	3.777	31.64
0 to 3.0 s (envelope closed)	0.06591	3.777	31.64

The response is bounded, not divergent, because no harmonic of the disturbance coincides exactly with f_n . The long run bound is $|\theta| = 0.06591$ rad and $|d\theta/dt| = 31.64$ rad/s.

4.2 Independent verification of the simulation output

Three checks confirm that the Simulink result is correct rather than a numerical artefact.

Check	Method	Expected	Simulink
Mean level	Static response to the reduced mean torque of 937.5 N·m: $937.5 / kt$	0.019267 rad	0.01927 rad

Check	Method	Expected	Simulink
Dominant frequency	Undamped natural frequency $\omega_n / 2\pi$	111.02 Hz	111.0 Hz
Long run bound	Sum of amplified harmonic amplitudes from Table 5.1	≤ 0.0724 rad	0.06591 rad

4.3 Engineering consequence

The peak twist is 3.2 times the static twist of 0.020552 rad. The peak torque carried by the shaft is $k_t \times 0.06591 = 3207 \text{ N}\cdot\text{m}$, which raises the peak shear stress at the shaft surface from 40.7 MPa to 130.7 MPa. The loading is also fully reversing at 111 Hz, which is 6.66 million stress cycles per hour. A single broken tooth has therefore turned a steady load into a high cycle fatigue problem.

5. Question 1.4 - Dominant torsional vibration frequency and resonance assessment

5.1 Method

The variable step solver returns time points that are not evenly spaced, so the logged $\theta(t)$ is first interpolated onto a uniform grid at 20 kHz. The static offset is removed, a Hann window is applied to suppress leakage, the record is zero padded to four times its length for finer frequency spacing, and the single sided amplitude spectrum is taken with fft.

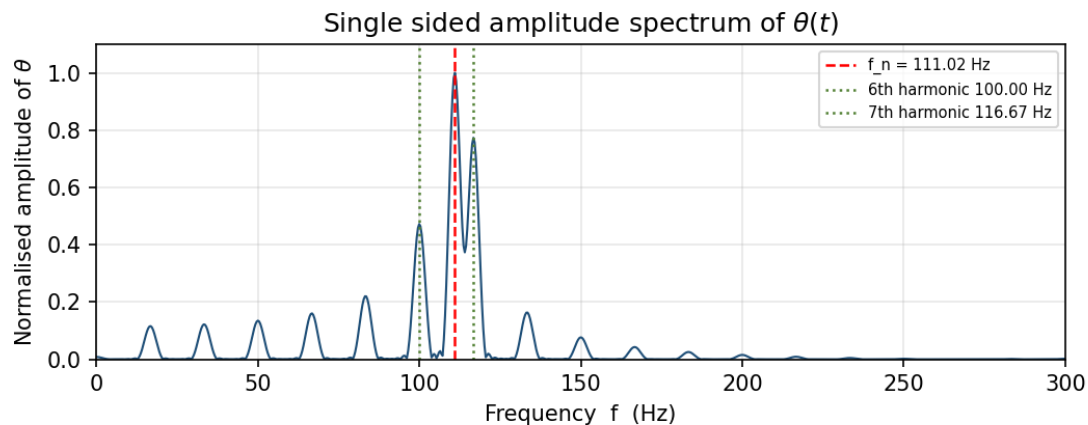


Figure 5: Amplitude spectrum of $\theta(t)$. The dominant peak sits at the natural frequency, with strong sidebands at the 6th and 7th harmonics of the once per revolution disturbance.

Answer to Question 1.4, part 1

The dominant torsional vibration frequency in the Simulink response is 111 Hz, which is the undamped natural frequency of the system, $f_n = \omega_n / 2\pi = 111.02$ Hz. The second and third peaks lie at 116.67 Hz and 100.00 Hz, which are the 7th and 6th harmonics of the 16.667 Hz once per revolution disturbance.

5.2 Why those frequencies appear

The disturbance is periodic at 16.667 Hz, so it acts on the system as a series of harmonics at multiples of that frequency. Expanding the rectangular torque gap as a Fourier series gives harmonic amplitudes $M_n = (2 M_0 / n\pi) |\sin(n\pi/16)|$. Each harmonic is then amplified by the standard undamped magnification factor $1 / |1 - r^2|$ with $r = n f_{rev} / f_n$.

Table 5.1 - Harmonic content of the disturbance and the resulting response

n	Frequency (Hz)	M _n (N·m)	$r = n f_{rev} / f_n$	$1/ 1-r^2 $	θ_n (rad)
1	16.67	124.20	0.1501	1.02	0.00261
2	33.33	121.81	0.3002	1.10	0.00275
3	50.00	117.90	0.4504	1.25	0.00304
4	66.67	112.54	0.6005	1.56	0.00362
5	83.33	105.87	0.7506	2.29	0.00498
6	100.00	98.03	0.9007	5.30	0.01068
7	116.67	89.20	1.0509	9.58	0.01757
8	133.33	79.58	1.2010	2.26	0.00370
9	150.00	69.38	1.3511	1.21	0.00173
10	166.67	58.82	1.5012	0.80	0.00096

The disturbance spreads its energy over many harmonics because the torque gap is narrow. Proximity to f_n then concentrates the response into the 6th and 7th harmonics, which is exactly where the spectrum in Figure 5 shows its strongest peaks.

5.3 Resonance assessment

Exact resonance requires a harmonic of the disturbance to coincide with the natural frequency, that is $n f_{rev} = f_n$ for some integer n . Here:

$$f_n / f_{rev} = 111.02 / 16.667 = 6.6611$$

This ratio is not an integer, so no harmonic falls exactly on f_n and the response stays bounded, as Section 4.1 confirms. That is the strict answer. It is not, however, the useful answer, because the 7th harmonic sits only 5.09 percent above f_n and is amplified by a factor of 9.58, while the 6th harmonic sits 9.93 percent below and is amplified by 5.30.

Answer to Question 1.4, part 2 - resonance assessment

Exact resonance does not occur, because $f_n / f_{rev} = 6.6611$ is not an integer and the response amplitude is therefore bounded rather than growing without limit.

The system is nevertheless in a severe near resonant condition. The 7th harmonic of the broken tooth disturbance falls at 116.67 Hz, within 5.09 percent of the 111.02 Hz natural frequency, and is magnified by 9.58. The result is a peak twist 3.2 times the static value and a peak shear stress of 130.7 MPa against a static 40.7 MPa. For design purposes this must be treated as a resonance problem.

5.4 Critical speeds

Solving $n f_{rev} = f_n$ for the running speed gives the speeds at which exact resonance would occur:

Harmonic n	Critical speed (rev/min)	Position relative to the 1000 rev/min operating speed
5	1332.2	Well clear above.
6	1110.2	The machine runs 9.93 % below this speed.
7	951.6	The machine runs only 5.09 % above this speed. This is the governing proximity.
8	832.6	Well clear below.

The operating speed of 1000 rev/min sits in the narrow window between two critical speeds, close to the lower one. The harmonics are spaced only 16.667 Hz apart near f_n , so there is no wide safe band at this speed.

5.5 Effect of the undamped assumption

Equation (1) contains no damping, so the 111.02 Hz free vibration excited by each torque interruption persists for the whole run and dominates the spectrum. Repeating the simulation with a light structural damping ratio of $\zeta = 0.02$, which is realistic for a steel shaft, reduces the peak by about 26 percent but leaves the twist at 2.36 times the static value. Damping alone does not solve the problem, which reinforces the recommendations below.

5.6 Recommendations

- **Replace the damaged gear.** This is the only measure that removes the disturbance at source. Everything else treats a symptom.
- **Do not run down through 951.6 rev/min under load.** Coasting through the 7th harmonic critical speed with the broken tooth still fitted would drive the system into exact resonance.
- **Retuning the shaft is not an effective standalone fix.** The harmonic spacing near f_n is only 16.667 Hz, so moving f_n clear of the significant harmonic content would need the natural frequency to exceed roughly 270 Hz, which means increasing the shaft diameter from 50 mm to about 78 mm or shortening it from 1.00 m to about 0.17 m. Neither is a light modification.
- **Monitor at 111 Hz and 116.7 Hz.** Growth of the 16.667 Hz once per revolution component and its harmonics is the diagnostic signature of a broken or cracked tooth and is what a condition monitoring system should be set to detect.

6. Summary of results

Question	Quantity	Result
Setup	Torsional stiffness, k_t	48 657.87 N·m/rad
Setup	Natural circular frequency, ω_n	697.55 rad/s
Setup	Natural frequency, f_n	111.02 Hz
Setup	Static twist before the break, $\theta(0)$	0.020552 rad
1.1	Disturbance model	Torque falls from 1000 to 0 N·m for 3.75 ms once every 60 ms (6.25 % duty, 16.667 Hz)
1.2	Response character	Beating oscillation about a reduced mean of 0.019267 rad, beat period about 0.18 s
1.3	Maximum angular displacement	0.06232 rad (3.571°) over 0 to 0.5 s; 0.06591 rad long run bound
1.3	Maximum angular velocity	29.13 rad/s over 0 to 0.5 s; 31.64 rad/s long run bound
1.4	Dominant vibration frequency	111 Hz, equal to the natural frequency f_n
1.4	Resonance assessment	Exact resonance does not occur ($f_n/f_{rev} = 6.6611$), but the system is in severe near resonance: the 7th harmonic at 116.67 Hz is within 5.09 % of f_n and is magnified 9.58 times, giving a peak twist 3.2 times static

7. Files in the submitted repository

File	Purpose
geared_shaft_model.slx	The Simulink model (Question 1.1)
sme3701_parameters.m	Defines all system parameters and prints the derived quantities. Run first.
build_geared_shaft_model.m	Builds and saves the Simulink model from source so it is fully traceable.
run_and_plot.m	Runs the model, produces the figures, and prints the answers to Questions 1.2, 1.3 and 1.4.
figures/	Exported plots at 300 dpi.

File	Purpose
screen_recording.mp4	Recording showing the model opened, configured and executed, with narration.